

Medicaid Expansion, Coverage Gains, and Mortality After Prison Release

ABSTRACT

Each year, more than 600,000 adults leave U.S. state prisons and enter a period of sharply elevated mortality risk. Medicaid expansion under the Affordable Care Act extended eligibility to most formerly incarcerated adults, but its post-release effects remain uncertain. Using staggered difference-in-differences and linked administrative data from the Census Bureau’s Justice Outcomes Explorer across twenty-three states, I estimate effects on Medicaid enrollment, mortality, employment, and earnings. Expansion produced large first-year Medicaid enrollment increases of 16 to 36 percentage points, depending on estimator. Average post-release mortality estimates were statistically indistinguishable from zero at the one-, three-, and five-year horizons. Corrected power calculations show minimum detectable effects of 0.209 percentage points at one year and 0.347 at five years. I can rule out large mortality reductions, but not effects as small as those found in general-population studies. I find no significant effects on employment or earnings. The strongest conclusion is therefore a large coverage gain paired with an informative but bounded average mortality null.

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The United States incarcerates more people per capita than any other nation, releasing approximately 600,000 adults from state prisons annually. These individuals face dramatically elevated mortality risk: drug overdose occurs at rates 129 times higher than in the general population during the first two weeks after release.¹ Medicaid expansion represents the largest expansion of health insurance coverage for justice-involved adults in U.S. history. Prior to the Affordable Care Act (ACA), most non-disabled adults without dependent children—the population profile of most formerly incarcerated people—were ineligible for Medicaid regardless of income. Between 2014 and 2023, forty states expanded eligibility to adults with incomes up to 138 percent of the federal poverty level.

The case for expansion reducing post-release mortality rests on two mechanisms: coverage enables access to medications for opioid use disorder (MOUD), which reduce overdose risk,² and facilitates treatment for chronic conditions prevalent among incarcerated populations. However, several countervailing forces may attenuate this effect: the post-release mortality spike is concentrated in the first two weeks before enrollment processes typically complete; the opioid epidemic intensified simultaneously; and many non-medical barriers to reentry—housing instability, unemployment, social isolation—are unaddressed by insurance.

A growing literature examines Medicaid and criminal justice. In the general

population, Miller, Johnson, and Wherry found that expansion reduced mortality by 9.4 percent among near-elderly adults.³ For justice-involved populations, Perera and colleagues compared Rhode Island (expansion) to North Carolina (non-expansion) and found reduced mortality among white formerly incarcerated individuals.⁴ Blumberger and colleagues found that Medicaid suspension alone did not reduce opioid overdose mortality.⁵ These studies are limited to one or two states.

This paper provides the first multi-state causal estimate of how Medicaid expansion affected post-release coverage, mortality, and socioeconomic reintegration. I exploit staggered adoption across states using estimators robust to heterogeneous treatment effects.^{6,7} My data come from the Census Bureau’s Justice Outcomes Explorer (JOE), which links criminal justice records from twenty-three states to mortality, Medicaid enrollment, and employment data.⁸

Study Data And Methods

My primary data source is the Justice Outcomes Explorer, a Census Bureau product developed with the University of Michigan’s Criminal Justice Administrative Records System (CJARS). CJARS links state criminal justice records to federal administrative data on mortality (Social Security Administration Numident), Medicaid enrollment (CMS MSIS/T-MSIS), and W-2 employment and earnings (IRS), producing aggregate statistics at the state-by-release-cohort-year-by-post-release-horizon level.

My unit of analysis is the state-by-release-cohort-year-by-post-release-horizon cell. Release cohorts span 2005 through 2019, with outcomes measured at one-, three-, and five-year horizons. I assign Medicaid expansion status and timing using Kaiser Family Foundation data.⁹ The analytic sample comprises 749 cells across twenty-three states (eighteen expansion, five never-treated controls: Florida, Georgia, Kansas, Texas, and Wisconsin).

I employ four estimators. My preferred approach is Local Projections Difference-in-Differences (LP-DiD),⁶ which compares newly treated to not-yet-treated and never-treated units, avoiding negative-weighting problems under staggered adoption. I supplement with standard two-way fixed effects (TWFE), TWFE with American Community Survey demographic controls, and TWFE using the Callaway-Sant’Anna group structure.⁷ All specifications cluster standard errors at the state level.

I assess parallel trends through event study specifications with leads and lags, conducting joint F-tests on pre-treatment coefficients. I compute minimum detectable effects (MDEs) at 80 percent power under clustered inference.

Limitations This study has several limitations. First, with twenty-three states and five controls, statistical power is constrained, though corrected power calculations show reasonably small MDEs (0.209 percentage points at one year, or 12.1 percent of baseline); this is roughly 1.6 times the Miller et al. general-population benchmark of 0.13 percentage points, so I can bound but not rule out

effects of that magnitude. Second, joint F-tests reject parallel pre-trends in mortality at the one- and three-year horizons, qualifying the causal interpretation; however, Rambachan-Roth honest difference-in-differences sensitivity analysis (online appendix) confirms the null is robust to trend violations up to twice the observed pre-treatment magnitude. Third, while I have incorporated CDC WONDER state-level overdose rates—which prove to be the most consequential control variable—I still lack unemployment rates and prison population controls. Fourth, aggregate data mask individual-level heterogeneity. Fifth, geographic assignment reflects state of charge rather than post-release residence. Sixth, the five never-treated control states include Wisconsin, whose BadgerCare program extended Medicaid coverage to adults up to 100 percent of the federal poverty level, making it a weaker counterfactual than the other four non-expansion controls.

Study Results

Medicaid enrollment. Expansion produced large, significant increases in post-release Medicaid enrollment (Exhibit 1). The TWFE estimate at one year is 36.0 percentage points ($p < 0.001$); LP-DiD yields 15.9 percentage points ($p = 0.046$). TWFE estimates remain positive at three and five years, but the preferred LP-DiD first stage attenuates sharply beyond one year. This attenuation appears to reflect a shrinking clean-control pool at longer horizons as more states adopt expansion, combined with cohort-composition shifts as earlier-treated cohorts exit the estimation window; I therefore anchor the first-stage claim on the one-year horizon. Event study analysis shows flat pre-trends for enrollment ($F = 1.57$, $p = 0.23$ at one year), supporting a causal interpretation.

Mortality. I find no statistically significant effect on post-release mortality across all estimators and horizons (Exhibit 1). The LP-DiD estimate at one year is -0.20 percentage points ($SE = 0.16$, $p = 0.23$). The 95 percent confidence interval $([-0.55, 0.15])$ rules out reductions larger than 0.55 percentage points, or roughly 35 percent of the pre-expansion mean of 1.72 percent. TWFE, TWFE with controls, and Callaway-Sant’Anna estimates are similarly null. The apparent positive +0.32 pp TWFE-with-controls estimate at one year reflects racial-composition adjustment in the absence of an overdose control; once the CDC WONDER overdose rate is included (Exhibit 4 Panel A), the coefficient returns to a small and statistically insignificant value (0.131 pp, $p = 0.380$), so I do not treat the controlled TWFE estimate as the preferred specification.

Power analysis. Corrected power calculations show MDEs of 0.209 percentage points at one year (12.1 percent of the pre-expansion mean), 0.267 percentage points at three years (8.4 percent), and 0.347 percentage points at five years (7.4 percent) (Exhibit 2). These are reasonably small relative to baseline mortality, though the general-population effect of 0.13 percentage points³ remains below my detection threshold at the one-year horizon. A formal two-one-sided-test (TOST) equivalence analysis (online appendix) cannot establish equivalence at the 0.13 pp Miller et al. margin at any horizon, but does establish equivalence at the MDE margin at the five-year horizon, consistent with an informative

bounded null.

Employment and earnings. No significant effects on W-2 employment (LP-DiD: 0.0 pp, $p = 1.00$ at one year) or earnings (LP-DiD: +\$265, $p = 0.61$).

Heterogeneity. Race-stratified analyses show null mortality effects across White (-0.06 pp, $p = 0.80$), Black (+0.00 pp, $p = 0.97$), and Hispanic (-0.46 pp, $p = 0.48$) subgroups, despite large Medicaid enrollment effects in all groups (Exhibit 3).

Heterogeneity by state overdose burden. When I augment the TWFE specification with CDC WONDER age-adjusted drug overdose mortality rates as a time-varying control, the overdose rate is a highly significant predictor of post-release mortality ($p = 0.001$ at one year). Interacting expansion status with a high-overdose indicator (above-median baseline overdose rate) yields a significant interaction term of 0.535 percentage points ($p < 0.001$), indicating that the mortality estimate is more favorable in low-overdose states (Exhibit 4). In split-sample estimation, expansion reduces one-year post-release mortality by 0.339 percentage points ($p = 0.018$) in low-overdose states, while the effect is null (0.104 pp, $p = 0.435$) in high-overdose states. Because this result is specific to the one-year horizon and depends on a control-augmented TWFE specification rather than the broader average-null pattern, I treat it as exploratory secondary evidence rather than as a co-equal headline finding.

Controls sensitivity. The state-level overdose rate emerges as the single most consequential control variable. Dropping it from the TWFE specification with controls shifts the one-year ATT from 0.131 ($p = 0.380$) to 0.319 ($p = 0.022$), underscoring the importance of accounting for the concurrent opioid epidemic when estimating expansion's mortality effects.

Robustness. Raw outcome trends by expansion wave show broadly parallel pre-period mortality and Medicaid enrollment paths across treated and never-treated states (online appendix). A treatment timing distribution figure confirms that treated-vs-never comparisons are well-supported across event-time (online appendix). A Goodman-Bacon decomposition shows that clean (treated-vs-never) comparisons receive 60.7 percent of the TWFE weight, and placebo tests using shifted treatment dates on the pre-expansion sample produce uniformly null estimates (online appendix). Results are robust to alternative treatment timing definitions, sample restrictions, and exclusion of states with partial CJARS coverage. Rambachan-Roth honest difference-in-differences sensitivity analysis confirms that the null is robust to allowing post-treatment trend violations up to twice the magnitude of observed pre-trends ($\bar{M} = 2.0$) at all horizons, substantially strengthening the null interpretation despite pre-trends concerns. Wild cluster bootstrap p-values closely track cluster-robust standard errors, confirming adequate inference with twenty-three clusters. A dose-response analysis finds a suggestive negative relationship between Medicaid enrollment gains and mortality reductions at the five-year horizon (slope significant at $p = 0.014$, $r = -0.947$), though the small number of states means I treat it as exploratory.

Measurement error from Census disclosure-protection rounding accounts for less than 1 percent of residual outcome variance. Oster bounds indicate that proportional selection on unobservables would need to be approximately six times larger than selection on observables to explain the controlled estimates.

Discussion

This study provides stronger evidence on coverage than on mortality. Medicaid expansion substantially increased coverage among formerly incarcerated adults, especially in the first post-release year, while average mortality estimates remained statistically indistinguishable from zero. This bounded null is more informative than a simple “failure to reject”: corrected power calculations show MDEs of 0.209 to 0.347 percentage points (12.1 to 7.4 percent of baseline), and the null survives Rambachan-Roth sensitivity analysis through $M = 2.0$, wild cluster bootstrap inference, and multiple alternative specifications. At the same time, I still cannot rule out effects of the magnitude found in general-population studies (0.13 percentage points),³ so the mortality evidence should be read as bounded rather than definitive. A suggestive dose-response relationship at the five-year horizon ($p = 0.014$) is better viewed as exploratory than as central evidence. With these considerations, several mechanisms could explain the average null.

The timing mismatch matters: post-release deaths concentrate in the first two weeks,^{1,10} while Medicaid enrollment typically requires days to weeks, so the highest-risk window often precedes active coverage.

The concurrent opioid epidemic appears to have offset protective effects, and my heterogeneity analysis is consistent with that possibility. When I control for CDC WONDER overdose rates and split the sample by baseline overdose burden, expansion appears more favorable in low-overdose states than in high-overdose states. But because this pattern comes from a narrower, control-sensitive analysis, I treat it as exploratory rather than as a definitive explanation of the average null. Even so, it reinforces the practical point that overdose context likely shapes whether coverage gains can translate into mortality improvements.

Most fundamentally, coverage may be necessary but, on its own, too slow or too limited to move the acute mortality margin I study. Insurance removes financial barriers but does not address provider unavailability, distrust, unstable housing, or competing survival needs. General-population studies document effects that accumulate over years;³ for formerly incarcerated adults, barriers to engagement may be sufficiently severe that coverage alone produces limited improvement.

This interpretation aligns with recent policy evolution. The Centers for Medicare and Medicaid Services’ Section 1115 reentry demonstration waivers—approved in eleven states as of 2024 with at least thirteen additional applications pending—allow Medicaid-covered pre-release services for up to ninety days before expected release.¹¹ By initiating treatment and care coordination before release, states can bridge the high-risk transition. The most defensible policy implication comes from the combination of a strong first-year enrollment result

and an honestly bounded average mortality null: Section 1115 reentry waivers are a promising next policy test, especially in high-risk overdose environments.

Conclusion

Medicaid expansion increased health insurance coverage among formerly incarcerated adults by sixteen to thirty-six percentage points in the first post-release year, but average post-release mortality estimates remained statistically indistinguishable from zero. The bounded average null is informative: corrected power calculations yield reasonably small minimum detectable effects (0.209 to 0.347 percentage points, or 7.4 to 12.1 percent of baseline), and the null is robust to Rambachan-Roth sensitivity analysis, wild cluster bootstrap inference, and multiple alternative specifications. At the same time, the study cannot rule out effects of the magnitude found in general-population research, so the mortality evidence should be read as bounded rather than definitive. The strong first-stage enrollment result confirms that expansion can reach this high-risk population. Exploratory subgroup and dose-response analyses suggest that overdose context may matter for whether coverage can translate into mortality improvements, but those results should remain secondary to the paper's main bounded-average conclusion. Evaluations of Section 1115 reentry demonstration waivers using individual-level data are well positioned to test whether pre-release services can turn enrollment gains into survival benefits.

EXHIBITS

Exhibit 1: Effect Of Medicaid Expansion On Post-Release Outcomes

Outcome	Horizon	TWFE	TWFE + Controls	LP-DiD (preferred)
Mortality (pp)	1-yr	-0.04 (0.13)	0.32* (0.13)	-0.20 (0.16)
	3-yr	0.07 (0.16)	0.23 (0.16)	-0.03 (0.20)
	5-yr	-0.04 (0.17)	-0.01 (0.17)	-0.21 (0.19)
Medicaid enrollment (pp)	1-yr	36.0** (6.8)	26.6** (4.8)	15.9* (7.2)
	3-yr	26.0** (5.8)	19.0** (2.7)	-0.5 (3.1)
	5-yr	21.5** (5.8)	17.0** (3.2)	1.1 (1.3)
W-2 em- ployment (pp)	1-yr	1.0 (1.7)	-0.4 (1.7)	0.0 (1.9)
W-2 earnings (\$)	1-yr	363 (332)	233 (332)	265 (505)

SOURCE Authors' analysis of Census Bureau Justice Outcomes Explorer data, 2005-2019 release cohorts. NOTES Standard errors clustered at the state level in parentheses. * $p < 0.05$, ** $p < 0.01$. TWFE = two-way fixed effects. LP-DiD = Local Projections Difference-in-Differences. Sample: 23 states (18 expansion, 5 never-treated). The "TWFE + Controls" column adjusts for ACS demographic controls; see Exhibit 4 Panel A for the preferred TWFE specification that additionally controls for the CDC WONDER state-level overdose rate.

Exhibit 2: Minimum Detectable Effects At 80 Percent Power (Corrected ICC)

Horizon	Pre-expansion mean (%)	MDE (pp)	MDE as % of mean
1-year	1.72	0.209	12.1
3-year	3.18	0.267	8.4
5-year	4.68	0.347	7.4

SOURCE Authors' calculations. NOTES 80 percent power, $\alpha = 0.05$ (two-sided), t-distribution with 22 degrees of freedom, corrected ICC estimation. MDE = minimum detectable effect.

Exhibit 3: Heterogeneity By Race, TWFE Estimates (1-Year Horizon)

Race	Mortality			Medicaid		
	ATT (pp)	SE	p-value	ATT (pp)	SE	p-value
White	-0.06	0.23	0.800	36.2	6.6	<0.001
Black	0.00	0.11	0.969	35.5	7.7	<0.001
Hispanic	-0.46	0.63	0.475	29.0	7.5	0.002

SOURCE Authors' analysis of Census Bureau Justice Outcomes Explorer data. NOTES TWFE estimates with state and cohort-year fixed effects. Standard errors clustered at the state level.

Exhibit 4: Overdose-Augmented Specifications and Heterogeneity By State Overdose Burden (1-Year Horizon)

Specification	Estimate (pp)	SE	p-value
Panel A: TWFE with Overdose Control			
Expansion ATT	0.131	0.14	0.380
Overdose rate (per 100k)	—	—	0.001
Panel B: Interaction Model			

Specification	Estimate (pp)	SE	p-value
Expansion (main effect)	-0.339	0.13	0.018
Expansion x High-overdose	0.535	0.12	<0.001
Panel C: Split-Sample Estimates			
Low-overdose states: Expansion ATT	-0.339	0.13	0.018
High-overdose states: Expansion ATT	0.104	0.13	0.435

SOURCE Authors’ analysis of Census Bureau Justice Outcomes Explorer data merged with CDC WONDER age-adjusted drug overdose mortality rates. NOTES TWFE estimates with state and cohort-year fixed effects. Standard errors clustered at the state level. High-overdose defined as above-median baseline (pre-expansion) age-adjusted drug overdose mortality rate. Results shown for the 1-year post-release horizon only; heterogeneity fades at 3- and 5-year horizons. The interaction term indicates that expansion’s mortality effect is 0.535 pp more favorable in low-overdose states.

ENDNOTES

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